

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Santhibastawad Road, Machhe, Belagavi-590018, Karnataka, India



MAJOR PROJECT PHASE -I

2022 SCHEME - BCS685

PROJECT REPORT

ON

“Privacy Masked Federated Learning for Heart Disease Diagnosis”

Submitted in the **partial fulfillment** of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

For the Academic Year 2025 - 2026

Group ID: 50

Name	USN	Signature
ABHISHEK R	1JS23CS004	
CHETAN KISHOR C G	1JS23CS032	
HONGIRAN D	1JS23CS062	
THIRUMANGAI S	1JS23CS180	

Signature of the Guide:

Signature of HOD:



Department of Computer Science and Engineering,
JSS Academy of Technical Education,
JSS Campus Rd, Dr. Vishnuvardhan Road, Bengaluru - 560060



JSS Academy of Technical Education
Department of Computer Science and Engineering

Major Project Phase - 1

Chapter No.	Title	Subtopics	Page No
1	Abstract	1.1 Problem Overview 1.2 Proposed Solution 1.3 Expected Outcome	
2	Introduction	2.1 Background 2.2 Need for the Project 2.3 Scope	
3	Problem Statement, Relevance & Objectives	3.1 Problem Statement 3.2 Importance of the Project 3.3 Objectives	
4	Literature Survey & Gap Analysis	4.1 Existing Systems/Research 4.2 Limitations of Existing Solutions 4.3 Gap Identified	
5	Proposed Methodology	5.1 Working of the Proposed System 5.2 Technologies/Tools Used 5.3 Workflow/Process	
6	System Design & Architecture	6.1 Architecture Diagram 6.2 Modules/Components 6.3 Database Design / DFD / UML Diagrams	
7	Technical Requirements & Constraints	7.1 Hardware Requirements 7.2 Software Requirements 7.3 Constraints and Limitations	
8	Project Planning & Timeline	8.1 Development Phases 8.2 Feasibility Analysis 8.3 Timeline / Gantt Chart	
9	Expected Outcomes	9.1 Expected Results 9.2 Benefits of the Project	
10	Implementation Details (Optional)	—	
11	Conclusion	11.1 Summary 11.2 Future Improvements	
12	References	12.1 Research Papers 12.2 Websites 12.3 Other Sources Used	

1. Abstract

1.1 Problem Overview

Heart disease remains one of the leading causes of mortality worldwide, and early diagnosis plays a critical role in reducing risks and improving patient outcomes. Existing AI-based diagnostic systems primarily rely on centralized data collection, where patient information from hospitals, laboratories, and wearable devices is stored and processed on centralized servers. This approach introduces major privacy and security concerns, as sensitive medical records, laboratory reports, ECG signals, and personal information are vulnerable to unauthorized access, misuse, or leakage. In recent years, the increasing use of generative AI tools by healthcare professionals has further amplified concerns regarding the accidental exposure of confidential patient data.

In addition to privacy issues, current diagnostic systems often focus only on static prediction models using limited datasets. Many systems do not effectively utilize multimodal medical information such as ECG data, clinical reports, symptoms, and lifestyle factors together. Existing models also lack transparency and actionable explanations, making it difficult for doctors and patients to understand why a prediction was generated.

Therefore, there is a need for a secure, privacy-preserving, explainable, and intelligent heart disease diagnosis system capable of collaborative learning without directly sharing sensitive patient data.

1.2 Proposed Solution

The proposed system, “Privacy Masked Federated Learning for Heart Disease Diagnosis,” introduces a secure and intelligent framework for heart disease prediction using Federated Learning and Privacy Masking techniques. Instead of sending raw patient data to a centralized server, the system trains AI models locally at hospitals or user devices and shares only model updates. This ensures that sensitive medical information never leaves its source.

The system integrates multiple advanced technologies including:

- Federated Learning for distributed and privacy-preserving training
- Named Entity Recognition (NER) based privacy masking to remove sensitive information from reports
- Transformer-based and BiLSTM models for multimodal data processing
- Counterfactual explainability methods for actionable insights
- Causal reasoning techniques to improve interpretability and trustworthiness

The system processes ECG data, clinical records, symptoms, and lifestyle information together to generate accurate and explainable predictions.

1.3 Expected Outcome

The expected outcome of this project is a secure AI-powered diagnostic platform capable of predicting heart disease while preserving patient privacy. The system aims to:

- Improve diagnostic accuracy using multimodal learning
- Prevent leakage of sensitive medical information
- Provide explainable predictions for doctors and patients
- Enable collaborative learning across hospitals without direct data sharing

The project is expected to demonstrate how privacy-preserving AI can be effectively applied in the healthcare domain while addressing real-world challenges in data security and medical intelligence.

2. Introduction

2.1 Background

Artificial Intelligence (AI) and Machine Learning (ML) have transformed various industries, including healthcare. In cardiology, AI-based systems are increasingly used to analyze ECG signals, patient records, medical reports, and physiological parameters to assist doctors in diagnosing heart diseases. These systems can process large volumes of medical data quickly and identify hidden patterns that may not be easily visible through manual analysis.

Heart disease diagnosis traditionally depends on clinical expertise, laboratory tests, ECG analysis, imaging reports, and patient history. However, the growing amount of healthcare data has created opportunities for AI systems to assist in predictive diagnosis and risk assessment.

Despite advancements in AI, existing diagnostic systems face several challenges. Most healthcare AI applications rely on centralized datasets where patient information is collected and stored in one location. This creates significant privacy risks and makes hospitals reluctant to share data due to ethical and legal concerns. Furthermore, many existing systems function as black-box models with limited interpretability.

Federated Learning has emerged as a promising solution to these problems. It allows multiple organizations to collaboratively train AI models without exchanging raw data. Combined with privacy masking and explainable AI techniques, Federated Learning can create secure and trustworthy healthcare systems.

2.2 Need for the Project

There is an increasing demand for secure and explainable AI systems in healthcare due to the following reasons:

- Rising incidents of medical data leakage and privacy violations
- Increased usage of online AI tools where patient reports may be exposed
- Hospitals being unwilling to share confidential medical records
- Lack of transparency in existing AI diagnostic systems
- Need for accurate and collaborative disease diagnosis using distributed data
- Need for real-time and personalized healthcare support

Current heart disease diagnosis systems often prioritize prediction accuracy while ignoring patient privacy and interpretability. Therefore, a system that combines privacy preservation, federated learning, multimodal analysis, and explainable AI is essential.

2.3 Scope

The scope of the proposed project includes:

- Development of a secure AI-based heart disease diagnosis system
- Integration of Federated Learning for decentralized model training
- Privacy masking of sensitive patient information using NER techniques
- Processing of multimodal data such as ECG signals, reports, symptoms, and lifestyle information
- Generation of explainable and interpretable predictions
- Development of doctor and patient dashboards
- Simulation of collaborative learning among multiple hospitals/devices

The project focuses primarily on heart disease diagnosis and privacy-preserving AI techniques within a healthcare environment.

3. Problem Statement, Relevance & Objectives

3.1 Problem Statement

Heart disease diagnosis systems often rely on centralized data processing, which exposes sensitive patient information and raises serious privacy concerns. In real-world practice, medical data such as lab reports and patient records are frequently shared across platforms or uploaded to AI tools, increasing the risk of data leakage and misuse. Additionally, hospitals are reluctant to share data, limiting the development of robust and generalizable diagnostic models. Therefore, there is a need for a secure, privacy-preserving approach that enables accurate heart disease diagnosis without compromising patient confidentiality.

3.2 Importance of the Project

The proposed project is important because it addresses both healthcare intelligence and patient privacy simultaneously. Medical data is highly sensitive, and unauthorized access or misuse can have severe ethical and legal consequences. By using Federated Learning and privacy masking techniques, the proposed system ensures that sensitive information remains protected.

The project also contributes to the advancement of explainable AI in healthcare. Instead of generating black-box predictions, the system provides understandable explanations and actionable insights. This improves trust among doctors and patients.

Key importance of the project includes:

- Protection of patient confidentiality

- Secure collaborative AI training
- Improved diagnostic support for doctors
- Explainable and interpretable predictions
- Real-world applicability in healthcare systems
- Contribution to ethical AI adoption in medicine

3.3 Objectives

The major objectives of the project are:

1. To develop an AI-based system for heart disease diagnosis.
2. To implement Federated Learning for privacy-preserving model training.
3. To apply privacy masking techniques using Named Entity Recognition.
4. To process multimodal healthcare data including ECG, reports, and lifestyle information.
5. To provide explainable predictions using counterfactual explanations.
6. To improve trust and transparency using causal reasoning.
7. To develop an interactive dashboard for doctors and patients.
8. To ensure that sensitive patient data is not directly shared or exposed.

4. Literature Survey & Gap Analysis

4.1 Existing Systems / Research

Paper 1: De-identification of Clinical Notes with Pseudo-Labeling using Regular Expression Rules and Pre-trained BERT (2025)

Research Focus

This research focuses on protecting sensitive patient information present in clinical notes using de-identification techniques based on regular expressions and BERT-based Named Entity Recognition (NER).

Approach and Methods Used

- Pre-trained BERT model
- Rule-based regular expression masking
- Pseudo-labeling techniques
- Clinical text de-identification
- OMOP medical dataset

Key Contributions

- Improved privacy masking for medical reports
- Combined rule-based and deep learning approaches
- Achieved better detection of sensitive information in clinical notes

Performance

- High de-identification performance on Korean clinical datasets
- Effective masking accuracy using limited annotated data

Practical or Theoretical

- Practical research implementation
- Focused mainly on clinical text privacy

Limitations

- Focuses only on textual de-identification
- No heart disease diagnosis system
- No multimodal learning
- No Federated Learning
- Does not provide explainable predictions

How Our System Improves It

Our system extends beyond simple text de-identification by integrating:

- Federated Learning
- Heart disease diagnosis
- ECG analysis
- Explainable AI
- Counterfactual reasoning
- Multimodal learning

Paper 2: Automatic De-identification of Textual Documents in Electronic Health Records (2010)

Research Focus

This paper reviews automated techniques for removing Protected Health Information (PHI) from healthcare documents.

Approach and Methods Used

- Rule-based de-identification
- Machine learning techniques
- HIPAA compliance methods
- Text anonymization

Key Contributions

- Early comprehensive study on healthcare data anonymization

- Discussed multiple PHI removal strategies
- Highlighted importance of healthcare privacy

Practical or Theoretical

- Mostly survey/review-based
- Not an end-to-end healthcare AI system

Limitations

- Traditional anonymization methods
- No AI-based diagnosis
- No deep learning integration
- No distributed learning
- No explainability

How Our System Improves It

Our project introduces:

- AI-powered diagnosis
- Federated privacy-preserving learning
- Modern NER-based masking
- Explainable and causal AI
- Real-time healthcare prediction

Paper 3: Precision at Heart – An IoT-based Vertical Federated Learning Approach for Cardiovascular Disease Risk Prediction (2026)

Research Focus

This paper proposes an IoT-enabled Vertical Federated Learning system for cardiovascular disease risk prediction using distributed healthcare data.

Approach and Methods Used

- Vertical Federated Learning (VFL)
- IoT healthcare devices
- Distributed machine learning
- Cardiovascular risk prediction

Key Contributions

- Demonstrated privacy-preserving cardiovascular prediction
- Utilized distributed feature sharing
- Integrated IoT with Federated Learning

Performance

- Achieved strong predictive accuracy for cardiovascular risk prediction
- Improved collaborative learning without direct data sharing

Practical or Theoretical

- Research-oriented practical implementation
- Mainly focused on risk prediction

Limitations

- Focused mainly on structured healthcare data
- Limited explainability
- No multimodal fusion
- No privacy masking layer

How Our System Improves It

Our project improves this work by adding:

- Privacy masking using NER
- Multimodal data fusion
- Explainable AI
- Causal reasoning
- What-if simulations
- Interactive doctor-patient dashboard

Paper 4: Heart Disease Detection using Novel Ensemble Approach (2025)

Research Focus

The paper focuses on improving heart disease detection accuracy using ensemble machine learning techniques.

Approach and Methods Used

- Ensemble learning
- Random Forest
- Decision Trees
- SVM
- Voting classifiers

Key Contributions

- Improved prediction accuracy through ensemble models
- Better classification compared to individual ML models

Performance

- High classification accuracy on benchmark datasets
- Strong tabular-data performance

Practical or Theoretical

- Practical machine learning implementation

Limitations

- Uses centralized datasets
- No patient privacy protection
- No multimodal learning
- No explainability
- Limited scalability

How Our System Improves It

Our system introduces:

- Federated Learning
- Privacy masking
- Deep multimodal learning
- ECG analysis
- Explainable AI
- Causal reasoning

Paper 5: Optimizing Heart Disease Diagnosis – A Reinforcement Learning-Based Ensemble Method (2025)

Research Focus

This paper uses reinforcement learning and ensemble methods to optimize heart disease diagnosis.

Approach and Methods Used

- Reinforcement learning
- Ensemble machine learning
- Learning automata
- Adaptive prediction systems

Key Contributions

- Improved prediction optimization
- Dynamic learning-based classification

Performance

- Improved classification metrics over conventional ML methods
- Good optimization capability

Practical or Theoretical

- Practical experimental implementation

Limitations

- No privacy-preserving mechanisms
- Centralized architecture
- No explainability
- No multimodal ECG-text fusion

How Our System Improves It

Our project adds:

- Privacy-preserving federated architecture
- Explainability using counterfactuals
- Multimodal processing
- Causal analysis
- Secure distributed learning

Paper 6: A Diagnosis Tool for Early Detection and Classification of Heart Disease using Transformer Mechanisms (2026)

Research Focus

This research proposes a transformer-based architecture for heart disease diagnosis and classification.

Approach and Methods Used

- Transformer mechanisms
- Deep learning
- Attention-based architectures
- Heart disease classification

Key Contributions

- Transformer-based heart disease prediction
- Improved feature representation
- Better performance than traditional ML models

Performance

- High prediction accuracy
- Strong classification performance

Practical or Theoretical

- Practical AI implementation

Limitations

- Centralized training approach
- No Federated Learning
- No privacy masking
- Limited explainability
- No causal reasoning

How Our System Improves It

Our project extends this research by integrating:

- Federated Learning
- Privacy masking
- Explainable AI
- Multimodal fusion
- Counterfactual analysis
- Causal inference

Paper 7: Data-driven Approaches for Early Diagnosis of Ischemic Heart Disease – Review of Recent Advancements (2026)

Research Focus

This paper reviews modern AI techniques for Ischemic Heart Disease diagnosis.

Approach and Methods Used

- Machine learning
- Deep learning
- Data-driven prediction
- ECG analysis

Key Contributions

- Comprehensive review of AI-based cardiac diagnosis
- Identified challenges and future opportunities

Practical or Theoretical

- Review paper
- Comparative analysis of existing systems

Limitations Identified by Authors

- Data privacy concerns
- Lack of explainability
- Limited generalization
- Insufficient multimodal integration
- Scalability issues

How Our System Addresses These Gaps

Our project directly addresses:

- Privacy preservation
- Explainable AI
- Federated collaborative learning
- Multimodal integration
- Personalized diagnosis

Paper 8: An Agentic AI System for Disease Diagnosis with Explanations (2026)

Research Focus

This paper introduces an Agentic AI framework for explainable disease diagnosis.

Approach and Methods Used

- Large Language Models (LLMs)
- Knowledge Graphs
- Retrieval-Augmented Generation (RAG)
- Multi-agent AI systems

Key Contributions

- Autonomous reasoning-based diagnosis
- Explainable conversational AI
- Knowledge-driven healthcare assistance

Performance

- Strong interpretability
- Improved reasoning capabilities

Practical or Theoretical

- Experimental research implementation

Limitations

- High computational complexity
- No Federated Learning
- No privacy masking
- Focused mainly on reasoning systems

How Our System Improves It

Our project combines:

- Privacy-preserving federated learning
- Lightweight multimodal diagnosis
- Explainability
- Real-world clinical applicability
- Secure distributed healthcare AI

Paper 9: Explainable AI for Automatic Heart Disease Diagnosis using 3DFMMecg Features (2026)

Research Focus

This paper proposes an explainable ECG-based heart disease diagnosis system.

Approach and Methods Used

- Explainable AI (XAI)
- ECG feature extraction
- 3DFMMecg representations
- Automatic diagnosis systems

Key Contributions

- Improved ECG interpretability
- Explainable prediction system
- ECG-focused diagnosis

Performance

- Strong ECG classification accuracy
- Better interpretability than black-box models

Practical or Theoretical

- Practical ECG-based implementation

Limitations

- Focused only on ECG signals
- No multimodal analysis
- No Federated Learning
- No privacy masking
- No causal reasoning

How Our System Improves It

Our system enhances this approach by introducing:

- Multimodal learning
- Federated Learning
- Privacy masking
- Counterfactual explanations
- Clinical text analysis
- Lifestyle-based personalization

4.2 Limitations of Existing Solutions

After analyzing multiple research papers and existing healthcare AI systems, several critical limitations were identified:

1. Centralized Data Dependency

Most existing systems rely on centralized storage and training methods where sensitive patient data is uploaded to servers. This creates:

- Privacy risks
- Data leakage concerns
- Security vulnerabilities
- Ethical and legal challenges

2. Lack of Privacy Preservation

Many systems focus primarily on prediction accuracy and ignore:

- Patient confidentiality
- Data anonymization
- Secure healthcare collaboration

- Privacy masking mechanisms

3. Limited Explainability

Traditional machine learning and deep learning systems behave like black-box models. Problems include:

- Lack of trust among doctors
- Poor interpretability
- No actionable insights
- Difficulty understanding predictions

4. Absence of Federated Learning

Most systems do not support distributed collaborative learning among hospitals or healthcare institutions. This results in:

- Poor model generalization
- Limited dataset diversity
- Reduced scalability

5. Single-Modality Analysis

Several existing systems rely only on:

- ECG data OR
- Clinical tabular data

Very few systems combine:

- ECG
- Reports
- Symptoms
- Lifestyle information

6. Lack of Causal Reasoning

Most AI systems only identify correlations and cannot explain:

- Why disease occurs
- Which factors truly influence risk
- How risk can be reduced

7. Limited Real-World Applicability

Many research systems remain:

- Theoretical
- Dataset-specific
- Non-deployable
- Computationally expensive

4.3 Gap Identified

The literature survey reveals a significant research gap in the development of a unified healthcare AI framework that simultaneously addresses:

- Privacy preservation
- Federated collaborative learning
- Explainable AI
- Multimodal medical analysis
- Causal reasoning
- Personalized healthcare intelligence

Existing systems focus only on isolated components such as:

- Prediction accuracy
- Explainability
- ECG analysis
- Privacy masking

However, no existing system effectively integrates all these technologies into a single secure and intelligent healthcare framework.

Research Gap Addressed by Our Project

The proposed system, “Privacy Masked Federated Learning for Heart Disease Diagnosis,” addresses these gaps through:

1. Federated Learning-Based Collaborative Diagnosis

- Enables hospitals to train models collaboratively
- No direct sharing of raw patient data
- Improves model generalization

2. Privacy Masking using NER

- Automatically masks sensitive patient information
- Prevents accidental data exposure
- Enhances healthcare data security

3. Multimodal AI Integration

The system combines:

- ECG signals
- Clinical reports
- Symptoms
- Lifestyle data

This improves diagnostic intelligence and personalization.

4. Explainable and Trustworthy AI

The system introduces:

- Counterfactual explanations
- Attention visualization
- Causal reasoning

This helps doctors understand predictions more effectively.

5. Real-World Healthcare Applicability

The project focuses on:

- Practical implementation
- Doctor-patient dashboards
- Privacy-first architecture
- Scalable deployment possibilities

6. Intelligent Personalized Diagnosis

The system supports:

- Risk prediction
- What-if simulations
- Personalized healthcare insights
- Decision support

Comparative Analysis Summary

The proposed project stands out because it combines:

- Federated Learning
- Privacy Masking

- Explainable AI
- Causal Reasoning
- Multimodal Learning
- Heart Disease Diagnosis

into a single integrated framework.

Unlike traditional systems that focus only on accuracy, the proposed system emphasizes:

- Security
- Trust
- Transparency
- Scalability
- Real-world healthcare usability

making it a next-generation intelligent healthcare solution.

Literature Survey Comparison Table

Year & Publication	Approach / Methods	Research Gap in Existing System	Our Project Enhancements	
2025 – De-identification using BERT	BERT, Regex Rules, NER	Only text masking, no diagnosis	Adds Federated Learning (FL), multimodal AI, and explainability	
2010 – EHR De-identification Review	Rule-based anonymization	Traditional methods, no AI diagnosis	Modern AI with privacy-preserving learning	
2026 – IoT-based Vertical Federated Learning for Cardiovascular Prediction	Vertical Federated Learning, IoT devices, distributed healthcare learning	No explainability, limited multimodal integration	Adds causal AI, multimodal fusion, and counterfactual explanations	
2025 – Ensemble Heart Disease Detection	Ensemble ML, Random Forest, SVM	Centralized architecture, no privacy protection	Adds federated privacy-preserving training	
2025 – Reinforcement Learning-based Heart Diagnosis	RL-based ensemble optimization	No explainability or privacy mechanisms	Adds explainable AI and secure learning	
2026 – Transformer-based Heart Disease Diagnosis	Transformer architectures, attention mechanisms	No Federated Learning or privacy layer	Adds privacy masking and distributed learning	
2026 – Data-driven Ischemic Heart Disease Review	Review of ML/DL methods	Highlights privacy and interpretability issues	Directly addresses identified gaps	
2026 – Agentic AI Diagnosis System	LLMs, RAG, Knowledge Graphs	High computational cost, no FL	Lightweight secure multimodal framework	
2026 – Explainable ECG Diagnosis using 3DFMMecg	Explainable ECG feature extraction	ECG-only approach, no privacy layer	Adds multimodal diagnosis and Federated Learning	
2025 – Federated Deep Learning for Secure Cardiac Prediction	Federated CNNs, distributed hospital training	No explainability, limited personalization	Adds explainable AI and personalized insights	
2026 – Privacy-Preserving Federated Healthcare Intelligence System	Federated Learning with encrypted aggregation	Focuses only on privacy, lacks multimodal diagnosis	Adds ECG, clinical, and lifestyle data fusion	
2025 – Edge-based Federated Learning for Smart Cardiology	Edge AI, Federated Learning, wearable integration	Limited causal reasoning and transparency	Adds causal inference and counterfactual analysis	

5. Proposed Methodology

5.1 Working of the Proposed System

The proposed system works by securely collecting and processing healthcare information from multiple sources while preserving patient privacy. The workflow begins when a doctor or patient uploads data such as ECG readings, clinical reports, symptoms, and lifestyle information.

The uploaded reports first pass through the Privacy Layer, where Named Entity Recognition (NER) techniques are used to detect and mask sensitive information such as names, addresses, IDs, and contact details. This ensures that personally identifiable information is removed before processing.

Next, the system extracts features from different modalities:

- ECG signals are processed using Time-Series Transformer or BiLSTM models.
- Clinical reports and textual information are processed using Transformer-based architectures.
- Structured tabular data is encoded using TabTransformer or FT-Transformer models.

After feature extraction, all modalities are combined using a multimodal fusion mechanism implemented through Graph Neural Networks (GNNs). This enables the system to analyze relationships between symptoms, ECG features, clinical indicators, and lifestyle factors.

The model training process is carried out using Federated Learning. Instead of transferring raw patient data to a central server, local devices or hospital nodes train the model independently and share only model parameters or updates. A central server aggregates these updates to improve the global model while ensuring that raw data never leaves the source.

The trained model then performs heart disease prediction and generates:

- Risk scores
- Diagnosis results
- Counterfactual explanations
- Personalized insights
- What-if simulations

The final output is displayed through an interactive dashboard for doctors and patients.

5.2 Technologies/Tools Used

The proposed system utilizes the following technologies and tools:

Programming Languages

- Python
- JavaScript

Frontend Technologies

- React.js
- HTML
- CSS
- Tailwind CSS

Backend Technologies

- FastAPI / Flask
- REST APIs

Machine Learning & AI Frameworks

- PyTorch
- TensorFlow
- Scikit-learn

Federated Learning Framework

- Flower Framework

Privacy & NLP Tools

- spaCy
- BERT-based NER

Explainability Tools

- DiCE (Counterfactual Explanations)
- Attention Visualization

Causal AI Tools

- DoWhy
- EconML

Database

- MongoDB / MySQL

Visualization Libraries

- Matplotlib
- Plotly
- Streamlit (optional)

5.3 Workflow/Process

The workflow of the proposed system consists of the following steps:

1. User uploads ECG data, reports, symptoms, and lifestyle details.
2. Sensitive information is detected and masked using NER.
3. Data is processed locally without external sharing.
4. Features are extracted using deep learning models.
5. Multimodal fusion combines information from different sources.
6. Local models are trained using Federated Learning.
7. Model updates are securely aggregated on the central server.
8. The global model is updated and redistributed.
9. Predictions and explanations are generated.
10. Results are displayed on the dashboard.

6. System Design & Architecture

6.1 Architecture Diagram

The architecture of the proposed system consists of multiple layers including:

1. Input Layer
 - ECG data
 - Lab reports
 - Symptoms
 - Lifestyle data
2. Privacy Layer
 - Named Entity Recognition
 - Data masking
 - Local processing
3. Feature Extraction Layer
 - BiLSTM / Time-Series Transformer
 - TabTransformer
 - Transformer models
4. Multimodal Fusion Layer
 - Graph Neural Network
5. Federated Learning Layer
 - Local client training

- Secure aggregation
- Global model update

6. Explainability & Causal Layer

- Counterfactual explanations
- Causal inference

7. Output Layer

- Disease diagnosis
- Risk score
- Recommendations
- What-if analysis

6.2 Modules/Components

The project contains the following modules:

1. User Authentication Module

- Doctor login
- Patient login
- Secure access management

2. Privacy Masking Module

- NER-based masking
- Anonymization
- Local processing

3. Data Processing Module

- ECG preprocessing
- Text preprocessing
- Tabular data encoding

4. Feature Extraction Module

- BiLSTM / Transformer models

5. Multimodal Fusion Module

- Graph Neural Network

6. Federated Learning Module

- Local training

- Weight aggregation
- Model synchronization

7. Explainability Module

- Counterfactual explanations
- Attention visualization

8. Causal Reasoning Module

- Cause-effect analysis

9. Dashboard Module

- Visualization
- Reports
- Risk graphs

6.3 Database Design / DFD / UML Diagrams

The system database stores:

- User credentials
- Masked patient data
- Model outputs
- Prediction history
- Federated model updates

Data Flow Diagram (DFD) includes:

- User input
- Privacy processing
- Feature extraction
- Federated training
- Prediction generation
- Dashboard output

UML components include:

- User class
- Doctor class
- Patient class
- AI model class
- Privacy module class
- Federated server class

7. Technical Requirements & Constraints

7.1 Hardware Requirements

Minimum hardware requirements:

- Processor: Intel i5 or higher
- RAM: 8 GB minimum
- Storage: 256 GB
- GPU: Optional but recommended for deep learning
- Internet connection for federated synchronization

7.2 Software Requirements

Software requirements include:

- Operating System: Windows/Linux
- Python 3.10+
- Node.js
- React.js
- FastAPI / Flask
- TensorFlow / PyTorch
- Flower Framework
- MongoDB / MySQL
- VS Code / PyCharm
- GitHub

7.3 Constraints and Limitations

The proposed project may face the following limitations:

- Federated Learning setup complexity
- Requirement of sufficient computational resources
- Limited access to real hospital datasets
- Simulated hospital environment instead of real deployment
- Time constraints for implementing advanced encryption methods
- Dependency on data quality and preprocessing

Despite these limitations, the project demonstrates the practical application of privacy-preserving AI in healthcare.

9. Expected Outcomes

9.1 Expected Results

The expected results of the project are:

- Accurate heart disease prediction system
- Secure and privacy-preserving AI workflow
- Successful implementation of Federated Learning
- Effective masking of sensitive patient information
- Explainable and interpretable predictions
- Interactive dashboard for doctors and patients
- Improved trust in AI-based healthcare systems

The project is expected to achieve good predictive performance while maintaining patient confidentiality.

9.2 Benefits of the Project

Benefits of the project include:

- Enhanced patient data privacy
- Reduced risk of medical data leakage
- Collaborative learning across distributed healthcare systems
- Explainable AI support for doctors
- Personalized healthcare insights
- Ethical and trustworthy AI adoption
- Better decision-making support in cardiology

11. Conclusion

11.1 Summary

The project “Privacy Masked Federated Learning for Heart Disease Diagnosis” proposes a secure and intelligent healthcare framework that combines Federated Learning, privacy masking, multimodal deep learning, explainable AI, and causal reasoning for heart disease diagnosis.

The system addresses critical real-world challenges such as medical data privacy, centralized AI risks, lack of interpretability, and limited collaboration among healthcare institutions. By ensuring that patient data remains local while enabling collaborative model training, the proposed approach enhances both privacy and diagnostic intelligence.

The integration of explainability and causal analysis further improves trustworthiness and usability for healthcare professionals.

11.2 Future Improvements

Future enhancements of the project may include:

- Real-time wearable device integration
- Advanced homomorphic encryption
- Blockchain-based medical data security

- Integration with hospital management systems
- Deployment on edge devices and mobile platforms
- More advanced causal reasoning models
- Real-world multi-hospital deployment
- Voice-based health interaction systems

The project has strong potential for expansion into a scalable and clinically deployable healthcare AI platform.

12. References

12.1 Research Papers

1. Research papers related to AI in heart disease diagnosis.
2. Papers on Federated Learning in healthcare.
3. Papers on explainable AI and causal inference.
4. Studies on privacy-preserving machine learning.
5. Literature on multimodal medical AI systems.

12.2 Websites

1. TensorFlow Official Documentation
2. PyTorch Official Documentation
3. Flower Federated Learning Documentation
4. Scikit-learn Documentation
5. spaCy Documentation
6. DoWhy Documentation
7. DiCE Documentation

12.3 Other Sources Used

1. Kaggle datasets
2. UCI Heart Disease Dataset
3. MIT-BIH ECG Dataset
4. Online technical articles and AI healthcare resources